

BSIS STUDENT COPY MATERIAL

OLAP SQL: ROLLUP, CUBE, and GROUPING SETS

1. OLAP Example 1 — ROLLUP by Year and Quarter

SQL Query

```
WITH SalesData AS
(
    SELECT
        YEAR(order_date) AS [Year],
        DATEPART(QUARTER, order_date) AS [Quarter],
        orders
    FROM sales_order
)
SELECT
    [Year],
    [Quarter],
    SUM(orders) AS [Orders]
FROM SalesData
GROUP BY ROLLUP ([Year], [Quarter])
ORDER BY [Year], [Quarter];
```

Overall Purpose

The query creates a summary of orders by year and quarter. It produces quarterly totals, yearly subtotals, and an overall grand total.

1.1 CTE (Common Table Expression)

WITH SalesData AS (...) creates a CTE. Think of it as a temporary result set prepared before the final report is produced. It is not a permanent database table.

The CTE prepares three pieces of information:

- Year extracted from order_date
- Quarter extracted from order_date
- The existing orders value

1.2 YEAR(order_date)

YEAR(order_date) extracts only the year portion of a date. For example, 2000-01-15 becomes 2000. AS [Year] gives the calculated column the name Year.

1.3 DATEPART(QUARTER, order_date)

DATEPART(QUARTER, order_date) extracts the quarter from the date:

Months	Quarter
January–March	1
April–June	2
July–September	3
October–December	4

In SQL Server, DATEPART(QUARTER, order_date) is used instead of a QUARTER(order_date) function.

1.4 SUM(orders)

SUM(orders) adds the order values in each group. For example, if the values are 87, 77, 91, and 125, their total is 380. COUNT(*) would count rows instead, giving 4.

1.5 GROUP BY ROLLUP

ROLLUP creates hierarchical subtotals and a grand total. For ROLLUP([Year], [Quarter]), the main levels are Year + Quarter, Year subtotal, and Grand Total.

Conceptual output:

Year	Quarter	Orders
2000	1	87
2000	2	77
2000	3	91
2000	4	125
2000	NULL	380
2001	1	139
2001	2	119
2001	3	10
2001	NULL	268
NULL	NULL	648

A NULL generated by ROLLUP can represent a subtotal level. For example, 2000 / NULL means the total for all quarters of 2000. NULL / NULL represents the grand total.

1.6 ORDER BY

ORDER BY [Year], [Quarter] controls the presentation order: first by year, then by quarter. In ascending SQL Server sorting, NULL values appear before non-NULL values, which can place the grand total at the top.

Mental Model

1. Stage 1 — Prepare: CTE extracts Year, Quarter, and Orders.
2. Stage 2 — Calculate: SUM(orders) calculates totals.
3. Stage 3 — Summarize: ROLLUP creates quarterly totals, yearly subtotals, and the grand total.
4. Stage 4 — Present: ORDER BY arranges the result.

One-line recitation answer

The query uses a CTE to extract the year and quarter from order_date, then uses SUM() and ROLLUP() to calculate quarterly totals, yearly subtotals, and the overall grand total of orders.

2. OLAP Example 2 — ROLLUP with Region and Quarter Filters

```
SELECT
    YEAR(order_date) AS [Year],
    DATEPART(QUARTER, order_date) AS [Quarter],
    region,
    SUM(orders) AS Orders
FROM sales_order
WHERE region IN ('South', 'Western')
    AND DATEPART(QUARTER, order_date) IN (1, 2)
GROUP BY ROLLUP
(
    YEAR(order_date),
    DATEPART(QUARTER, order_date),
```

```

        region
    )
ORDER BY
    [Year],
    [Quarter],
    region;

```

Overall Purpose

The query summarizes total orders only for the South and Western regions and only for Q1 and Q2. ROLLUP then creates detailed totals and hierarchical subtotals.

2.1 SELECT and Calculated Columns

The final result displays Year, Quarter, Region, and Orders. YEAR(order_date) extracts the year; DATEPART(QUARTER, order_date) extracts the quarter; region is taken directly from the table; SUM(orders) calculates the total order value for each group.

2.2 WHERE Filters

WHERE region IN ('South', 'Western') keeps only South and Western records.

This is equivalent to:

```
WHERE region = 'South' OR region = 'Western'
```

AND DATEPART(QUARTER, order_date) IN (1, 2) keeps Q1 and Q2 only.

Because AND is used, both conditions must be true. For example, South + Q1 is included, South + Q3 is excluded, and North + Q1 is excluded.

2.3 ROLLUP Hierarchy

The hierarchy is:

- Year
- Quarter
- Region

Level 1: Year + Quarter + Region — the most detailed totals.

Level 2: Year + Quarter — region is rolled up. For example, 2000 / 1 / NULL means the Q1 total for all selected regions in 2000.

Level 3: Year — quarter and region are rolled up. For example, 2000 / NULL / NULL is the total for the selected quarters and selected regions in 2000.

Level 4: Grand Total — NULL / NULL / NULL is the total of all records that passed the WHERE filter.

2.4 Important Rule: WHERE Happens Before ROLLUP

The query first filters the data and only then performs the grouping and rollup. Therefore, the grand total is not the total of the entire sales_order table. It is the total for South + Western and Q1 + Q2 only.

Quick Exam Review

SQL Part	Meaning
YEAR(order_date)	Extracts the year
DATEPART(QUARTER, order_date)	Extracts the quarter
region	Provides the sales region

SUM(orders)	Adds order values
FROM sales_order	Specifies the source table
WHERE	Filters records
IN ('South','Western')	Keeps South and Western
IN (1,2)	Keeps Q1 and Q2
GROUP BY	Groups records
ROLLUP	Creates subtotals and a grand total
ORDER BY	Sorts the final output

Easy memory guide:

WHERE = FILTER

GROUP BY = GROUP

SUM = TOTAL

ROLLUP = SUBTOTAL + GRAND TOTAL

ORDER BY = SORT

3. OLAP Example — CUBE Operation

```
WITH SalesData AS
(
    SELECT
        YEAR(order_date) AS [Year],
        DATEPART(QUARTER, order_date) AS [Quarter],
        orders
    FROM sales_order
)
SELECT
    [Year],
    [Quarter],
    SUM(orders) AS [Orders]
FROM SalesData
GROUP BY CUBE ([Year], [Quarter])
ORDER BY
    CASE
        WHEN [Year] IS NULL AND [Quarter] IS NULL THEN 1
        WHEN [Year] IS NULL AND [Quarter] IS NOT NULL THEN 2
        WHEN [Year] IS NOT NULL AND [Quarter] IS NULL THEN 3
        ELSE 4
    END,
    [Year],
    [Quarter];
```

Overall Purpose

The query calculates total orders per year and quarter and also produces year subtotals, quarter subtotals across all years, and a grand total.

3.1 CUBE

GROUP BY CUBE([Year], [Quarter]) generates all possible grouping combinations for the two dimensions.

Grouping	Meaning
Year + Quarter	Orders for a specific year and quarter
Year only	Total orders for that year across all quarters
Quarter only	Total orders for that quarter across all years

Neither Year nor Quarter	Grand total of all orders
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3.2 Meaning of NULL in CUBE

NULL does not necessarily mean missing data. In CUBE output, it can indicate that a dimension has been rolled up into a subtotal or total.

Examples:

- 2024 / NULL = total orders for all quarters of 2024.
- NULL / Q1 = total orders for Q1 across all years.
- NULL / NULL = grand total of all orders.

3.3 Why ORDER BY CASE Is Used

The CASE expression assigns a display priority so that the CUBE results appear in a deliberate order:

CASE Priority	Rows Classified
1	Year NULL + Quarter NULL = Grand Total
2	Year NULL + Quarter not NULL = Quarter totals across all years
3	Year not NULL + Quarter NULL = Year totals
4	All other rows = Detailed Year + Quarter records

Professor's Mental Model

Think of CUBE as producing several related reports in one query:

- Year + Quarter details
- Year totals
- Quarter totals across all years
- Grand total

This is useful in business intelligence, reporting, and decision-support systems because one query can provide a multidimensional summary.

Key exam keyword: CUBE = all combinations of grouping dimensions + subtotals + grand total.

One-sentence explanation

The CTE prepares the Year and Quarter data, SUM() calculates total orders, CUBE generates detailed records plus subtotals and a grand total, and ORDER BY CASE controls how those results are displayed.

4. OLAP Example — GROUPING SETS / ROLLUP Operation

```
WITH SalesData AS
(
    SELECT
        YEAR(order_date) AS [Year],
        DATEPART(QUARTER, order_date) AS [Quarter],
        orders
    FROM sales_order
)
SELECT
    [Year],
    [Quarter],
    SUM(orders) AS [Orders]
FROM SalesData
```

```

GROUP BY
    [Year],
    ROLLUP ([Quarter])
HAVING [Year] IS NOT NULL
ORDER BY
    [Year],
    [Quarter];

```

Overall Purpose

The query produces quarterly details and yearly subtotals, but removes the overall grand total.

4.1 CTE Preparation

The CTE extracts Year, Quarter, and Orders from sales_order. It serves as the preparation stage before the summary is calculated.

4.2 GROUP BY [Year], ROLLUP([Quarter])

This creates a hierarchy in which each year contains its quarters, followed by a subtotal for that year.

Conceptually:

- 2000 → Q1, Q2, Q3, Q4 → 2000 subtotal
- 2001 → Q1, Q2, Q3 → 2001 subtotal

4.3 ROLLUP vs. CUBE

Operation	Produces	Use in This Example
ROLLUP([Quarter]) within Year	Quarter details + Year subtotal	Matches the required output
CUBE([Year],[Quarter])	Year+Quarter, Year totals, Quarter totals, Grand Total	Produces additional combinations that are not needed here

4.4 HAVING [Year] IS NOT NULL

HAVING [Year] IS NOT NULL removes rows where Year is NULL. In this ROLLUP structure, that removes the overall grand total while keeping the yearly subtotals and quarterly details.

The intended result therefore includes:

- 2000 / NULL / 380 — Year 2000 subtotal
- 2000 / 1 / 87 — Q1 detail
- 2000 / 2 / 77 — Q2 detail
- 2000 / 3 / 91 — Q3 detail
- 2000 / 4 / 125 — Q4 detail
- 2001 / NULL / 268 — Year 2001 subtotal
- 2001 / 1 / 139 — Q1 detail
- 2001 / 2 / 119 — Q2 detail
- 2001 / 3 / 10 — Q3 detail

4.5 ORDER BY

ORDER BY [Year], [Quarter] organizes the output chronologically by year and then by quarter.

Final BSIS Exam Memory Guide

Concept	Remember	Typical Result
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CTE	Preparation stage	Creates SalesData
YEAR()	Extracts year	2024
DATEPART(QUARTER, ...)	Extracts quarter	1–4
SUM()	Calculates totals	87 + 77 = 164
WHERE	Filters before grouping	South/Western; Q1/Q2
ROLLUP	Details → subtotal → higher-level total	Year+Quarter → Year → Grand Total
CUBE	All combinations of grouping dimensions	Year+Quarter, Year, Quarter, Grand Total
HAVING	Filters grouped/aggregated results	Can remove the grand total
ORDER BY	Sorts final output	Year → Quarter → Region

Fast Recitation Answers

What does a CTE do?

It prepares a temporary result set that can be used by the main query.

What does YEAR(order_date) do?

It extracts the year from order_date.

What does DATEPART(QUARTER, order_date) do?

It extracts the quarter from order_date.

Why use SUM(orders) instead of COUNT(*)?

SUM adds the order values; COUNT(*) counts rows.

What does WHERE do?

It filters records before grouping.

What does ROLLUP do?

It creates hierarchical subtotals and higher-level totals.

What does CUBE do?

It creates all grouping combinations for the specified dimensions, including subtotals and grand total.

What can NULL mean in ROLLUP/CUBE output?

It can indicate a subtotal or total level rather than missing data.

Why use HAVING [Year] IS NOT NULL in Example 4?

It removes the grand-total row while retaining yearly subtotals and quarterly details.

What does ORDER BY do?

It controls the order in which the final rows are displayed.